Methods in Health Data Science with R

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1 Overview

This document regroups the four intensive courses into one coherent syllabus:

1. **BIOSTATR** - Applied Biostatistics with R (4 weeks - 24h)
2. **ML4HOR** - Machine Learning for Health Outcomes with R (4 weeks - 24h)
3. **SURV4CR** - Survival Analysis for Clinical Research with R (4 weeks - 24h)
4. **METAR** - Conducting Meta-Analysis with R (4 weeks - 24h)

1.1 Practical informations

- **Instructor**: Imad EL BADISY.
- Each course follows the same training structure: 4 weeks, 12 sessions, 2 hours per session, for a total of 24 hours.
- Sessions are scheduled on Monday, Wednesday, and Friday, 17:00-19:00 Morocco time.
- Before each session, participants receive the slides, self-contained notebooks, datasets, R scripts, and exercises. Exercises are reviewed and corrected during the sessions.
- The teaching language is French, with English for materials of statistical and methodological terminology.
- The four courses are designed to be taken independently.
- No prerequisite is mandatory for any single course. The required foundations are introduced progressively.

NB: For participants who want to build comprehensive mastery of the main methods in health data science, the proposed order is recommended.

1.2 Programme design

The programme is designed as a progressive health data science pathway. `BIOSTATR` establishes the core statistical and reproducible R workflow. `ML4HOR` introduces prediction, validation, interpretation, and patient phenotyping. `SURV4CR` extends health-data modelling to time-to-event outcomes. `METAR` closes the pathway with evidence synthesis and pooled inference.

2 Shared pedagogical format

Each session combines methodological concepts, live R demonstration, guided practical work, interpretation discussion, and reproducible reporting. Across all four courses, participants work with R, RStudio, and Quarto so that statistical reasoning, code, tables, figures, and interpretation remain integrated.

Assessment is practical. Each course ends with a short reproducible Quarto report or mini-project adapted to the course topic.

3 BIOSTATR

Applied Biostatistics with R

3.1 Course description

`BIOSTATR` provides a practical and rigorous introduction to applied biostatistics for health research using R. It covers the full analytical workflow: study questions, data types, descriptive statistics, visualization, inference, regression models, survival analysis, diagnostic accuracy, missing data, causal thinking, and reproducible reporting with Quarto.

The course emphasizes estimation, interpretation, reproducibility, and appropriate use of statistical methods in clinical research.

3.2 Learning objectives

By the end of `BIOSTATR`, participants will be able to:

1. formulate health research questions in statistical terms;
2. identify study designs and variable types;
3. organize reproducible analyses using R and Quarto;
4. summarize and visualize health data appropriately;
5. interpret confidence intervals, p-values, and effect estimates;
6. compare groups using suitable statistical tests;
7. fit and interpret linear and logistic regression models;
8. understand the basics of survival analysis and diagnostic accuracy;
9. recognize issues related to missing data and repeated measures;
10. produce a reproducible statistical report.

3.3 Detailed schedule

Table 1: `BIOSTATR` detailed schedule

No.	Topic	Main Content
1	Introduction to Biostatistics and Study Design	Role of biostatistics, study designs, data types, bias, confounding, research questions
2	R and Reproducible Workflows	R basics, RStudio projects, data import, project structure, Quarto workflow
3	Descriptive Statistics	Central tendency, dispersion, frequencies, missing values, Table 1
4	Data Visualization	Distribution plots, group comparisons, uncertainty visualization, publication-ready figures
5	Estimation and Statistical Inference	Sampling, confidence intervals, p-values, effect sizes, clinical interpretation

No.	Topic	Main Content
6	Group Comparisons	t-test, Wilcoxon test, chi-square test, Fisher test, ANOVA, Kruskal-Wallis
7	Correlation and Linear Regression	Pearson/Spearman correlation, simple and multiple linear regression, diagnostics
8	Logistic Regression and GLMs	Binary outcomes, odds ratios, adjusted models, discrimination, calibration
9	Sample Size and Reporting	Power, precision, study planning, statistical analysis plan, reporting guidelines
10	Survival Analysis and Diagnostic Accuracy	Kaplan-Meier, censoring, Cox model, sensitivity, specificity, PPV, NPV, ROC curves
11	Missing Data, Longitudinal Data and Causal Thinking	MCAR/MAR/MNAR, multiple imputation, repeated measures, DAGs, confounding
12	Final Synthesis and Mini-Project	Complete analysis workflow, interpretation, Quarto report, participant presentations

4 ML4HOR

Machine Learning for Health Outcomes with R

4.1 Course description

ML4HOR provides a practical introduction to machine learning for health outcomes research using R. The course covers the complete supervised learning workflow and explore clustering as a final extension for patient phenotyping, subgroup discovery, and exploratory health-data analysis.

The course emphasizes validation, data leakage prevention, appropriate performance metrics, calibration, interpretability, clinical relevance, and reproducible reporting.

4.2 Learning objectives

By the end of `ML4HOR`, participants will be able to:

1. formulate machine learning questions in health outcomes research;
2. prepare tabular health datasets for ML analysis;
3. fit supervised learners;
4. evaluate models using resampling-based validation;
5. tune and compare learners fairly;
6. assess discrimination, calibration, and clinical usefulness;
7. interpret ML models using global and local explanation methods;
8. apply clustering for patient phenotyping and subgroup discovery;
9. distinguish prediction, clustering, and causal estimation;
10. produce a reproducible ML report with Quarto.

4.3 Detailed schedule

Table 2: ML4HOR detailed schedule

No.	Topic	Main Content
1	Introduction to ML for Health Outcomes	Prediction vs explanation, health outcome types, supervised vs unsupervised learning, target definition, leakage risks
2	R Workflow and Data Preparation	R project structure, Quarto, preprocessing, missing data, encoding, scaling, analysis-ready datasets
3	Resampling and Performance Metrics	Train/test split, cross-validation, bootstrap, RMSE, MAE, AUC, F1, log-loss, Brier score
4	Linear and Regularized Models	GLM, logistic regression, ridge, lasso, elastic net, shrinkage, high-dimensional predictors
5	Tree-Based Models and Random Forests	CART, pruning, random forests, out-of-bag error, variable importance, overfitting
6	Boosting, Kernels and Ensemble Methods	XGBoost, LightGBM, SVM, kNN, stacking, Super Learner
7	Hyperparameter Tuning and Model Selection	Grid search, random search, nested cross-validation, optimism, model selection bias
8	Learner Comparison and Benchmarking	Fair comparison, shared resampling, benchmark tables, uncertainty in rankings, reporting model comparisons
9	Calibration and Clinical Utility	Calibration curves, discrimination vs calibration, thresholds, sensitivity, specificity, PPV, NPV, ROC-AUC
10	Model Interpretation and Explainability	Permutation importance, PDP, ICE, ALE, LIME, SHAP, global vs local explanations
11	Clustering and Patient Phenotyping	k-means, hierarchical clustering, PAM, Gaussian mixtures, distance metrics, number of clusters, cluster profiling
12	End-to-End ML Pipeline and Mini-Project	Full workflow: prepare, fit or cluster, evaluate, tune/compare, calibrate, interpret, report

5 SURV4CR

Survival Analysis for Clinical Research with R

5.1 Course description

SURV4CR provides a practical and methodologically rigorous introduction to survival analysis for clinical research using R. It covers the full time-to-event analysis workflow: survival data structure, censoring, Kaplan-Meier estimation, log-rank tests, Cox regression, proportional hazards diagnostics, competing risks, RMST, survival machine learning, dynamic RMST analysis, and temporal phenotyping.

The course emphasizes clinical interpretation, correct handling of censoring, model assumptions, reproducible analysis, and responsible use of machine learning for time-to-event outcomes.

5.2 Learning objectives

By the end of `SURV4CR`, participants will be able to:

1. define time origin, event, follow-up time, and censoring mechanism;
2. structure survival data using R;
3. estimate and interpret Kaplan-Meier curves;
4. perform and interpret log-rank tests;
5. fit and assess Cox proportional hazards models;
6. diagnose and handle non-proportional hazards;
7. analyze competing risks using cumulative incidence and Fine-Gray models;
8. use RMST as a clinically interpretable survival estimand;
9. fit and evaluate survival machine learning models;
10. interpret survival ML predictions;
11. compute dynamic RMST curves;
12. apply temporal clustering for survival phenotyping;
13. produce a reproducible survival analysis report.

5.3 Detailed schedule

Table 3: `SURV4CR` detailed schedule

No.	Topic	Main Content
1	Survival Data Structure and Study Design	Time-to-event outcomes, time origin, event definition, right censoring, independent censoring, `Surv()` objects
2	Kaplan-Meier, Nelson-Aalen and Log-Rank Test	Kaplan-Meier estimator, cumulative hazard, survival curves, median survival, log-rank test, RMST introduction
3	Cox Proportional Hazards Model	Cox model, hazard ratios, partial likelihood, adjusted models, PH assumption, Schoenfeld residuals
4	Cox Model Extensions	Stratified Cox models, time-varying covariates, time-varying effects, frailty models, clustered survival data
5	Competing Risks and Flexible Parametric Models	Cause-specific hazards, cumulative incidence, Fine-Gray model, flexible parametric survival models, cure models overview
6	RMST as a Clinical Estimand	RMST definition, RMST difference and ratio, choice of time horizon, interpretation under non-proportional hazards
7	Survival Machine Learning Foundations	Limits of Cox models, survival prediction matrices, penalized Cox, random survival forests, oblique forests
8	Survival ML Evaluation and Tuning	IPCW metrics, C-index, Brier score, integrated Brier score, time-dependent AUC, cross-validation, tuning

No.	Topic	Main Content
9	Survival ML Interpretation	Variable importance, PDP, ICE, ALE, SHAP, local explanations, clinical interpretation of survival predictions
10	Dynamic RMST with `tvrmst`	RMST as a function of time, dynamic RMST curves, treatment contrast curves, bootstrap confidence intervals
11	Temporal Phenotyping with `unsurv`	Clustering predicted survival curves, functional distances, cluster medoids, risk phenotypes, cluster-level RMST
12	End-to-End Survival Analysis Project	Complete workflow: KM, Cox, RMST, survival ML, interpretation, dynamic RMST, phenotyping

6 METAR

Conducting Meta-Analysis with R

6.1 Course description

METAR provides a practical and methodologically rigorous introduction to conducting meta-analysis with R. It covers the complete evidence-synthesis workflow: formulating a review question, developing an analysis-ready extraction table, computing effect sizes, fitting fixed-effect and random-effects models, assessing heterogeneity, producing forest plots, exploring moderators, assessing publication bias, evaluating certainty of evidence, and preparing a reproducible Quarto report.

The course emphasizes transparent reporting, correct interpretation of pooled estimates, reproducible analysis, and publication-quality tables and figures.

6.2 Learning objectives

By the end of `METAR`, participants will be able to:

1. formulate a meta-analysis question using the PICO framework;
2. structure extracted study data for analysis in R;
3. compute effect sizes for binary and continuous outcomes;
4. fit fixed-effect and random-effects meta-analysis models;
5. interpret pooled estimates, confidence intervals, and prediction intervals;
6. quantify and explore heterogeneity using Q, I^2, and tau^2;
7. produce publication-quality forest plots;
8. conduct subgroup analysis and meta-regression;
9. assess small-study effects and publication bias;
10. perform sensitivity and influence analyses;
11. summarize risk of bias and certainty of evidence;
12. produce a reproducible meta-analysis report with Quarto.

6.3 Detailed schedule

Table 4: `METAR` detailed schedule

No.	Topic	Main Content
1	Introduction to Systematic Reviews and Meta-Analysis	Evidence hierarchy, narrative review, systematic review, meta-analysis, PICO, PRISMA workflow, pooled estimates, heterogeneity
2	Protocol, Search Strategy and Study Selection	Review protocol, eligibility criteria, databases, search strategy, screening workflow, PRISMA flow diagram
3	Data Extraction and Risk of Bias	Extraction tables, study characteristics, outcomes, sample sizes, event counts, means, SDs, CIs, risk-of-bias variables, `robvis`
4	Preparing Extracted Data in R	Importing Excel/CSV files, data cleaning, unit harmonization, zero cells, missing standard errors, consistency checks
5	Effect Sizes for Binary Outcomes	Odds ratio, risk ratio, risk difference, log transformation, standard errors, continuity correction
6	Effect Sizes for Continuous Outcomes	Mean difference, standardized mean difference, Hedges' g, transformation of reported statistics
7	Fixed-Effect and Random-Effects Models	Inverse-variance weighting, common-effect model, random-effects model, DerSimonian-Laird, REML, Paule-Mandel, prediction intervals
8	Heterogeneity and Sensitivity Analysis	Cochran's Q, I^2, tau^2, heterogeneity CIs, Baujat plots, leave-one-out analysis, influence diagnostics
9	Forest Plots and Result Interpretation	Forest plot anatomy, study weights, summary diamond, prediction interval, scale choice, subgroup forest plots
10	Subgroup Analysis and Meta-Regression	Pre-specified subgroups, between-subgroup tests, study-level moderators, meta-regression, bubble plots, ecological fallacy
11	Publication Bias and Certainty of Evidence	Funnel plots, contour-enhanced funnel plots, Egger test, Begg test, Peters test, trim-and-fill, small-study effects, GRADE overview
12	Reproducible Reporting and Final Project	Quarto report, PRISMA diagram, risk-of-bias plot, effect-size table, forest plot, heterogeneity, publication bias

7 Overall expected outcome

At the end of the four-course pathway, participants will be able to conduct reproducible health data analyses in R across common research settings: general biostatistics, machine learning for health outcomes, clinical survival outcomes, and evidence synthesis.